CMAN — Manual INT8 Symmetric Quantization

(a) Scale factor

Max |W|: 2.31
S = max(|W|) / 127: 0.0181889763779528

Original W (FP32):

0.85	-1.20	0.34	2.10
-0.07	0.91	-1.88	0.12
1.55	0.03	-0.44	-2.31
-0.18	1.03	0.77	0.55

(b) Quantized W_q (INT8, clamped to [-128,127])

W_q:

47	-66	19	115
-4	50	-103	7
85	2	-24	-127
-10	57	42	30

(c) Dequantized W_deq = W_q × S

W_deq:

0.85	-1.20	0.35	2.09
-0.07	0.91	-1.87	0.13
1.55	0.04	-0.44	-2.31
-0.18	1.04	0.76	0.55

(d) Absolute error |W - W_deq|

Error (Max):

0.005	0.000	0.006	0.008
0.003	0.001	0.007	0.007
0.004	0.006	0.003	0.000
0.002	0.007	0.006	0.004

Largest-error element: **row 1, col 4 (W=2.10)**
Largest abs error: **0.008268**
MAE (mean abs error): **0.004326**

(e) Bad scale experiment: S_bad = 0.01

W_q (bad, clamped):

85	-120	34	127
-7	91	-128	12
127	3	-44	-128
-18	103	77	55

W_deq (bad):

0.85	-1.20	0.34	1.27
-0.07	0.91	-1.28	0.12
1.27	0.03	-0.44	-1.28
-0.18	1.03	0.77	0.55

Abs error (bad):

0.00	0.00	0.00	0.83
0.00	0.00	0.60	0.00
0.28	0.00	0.00	1.03
0.00	0.00	0.00	0.00

MAE (bad): **0.171250**

Explanation: **When S is too small, large weights exceed INT8 range [-128,127] and get clamped, causing severe saturation errors at the largest-magnitude weights.**